

ORAL CONTRACEPTIVES

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- **Definition :** Drugs taken orally (pills) to prevent pregnancy by inhibiting the maturation of follicles and preventing ovulation.
- **Effect on Menstrual Cycle :** Alters the normal cycle into an **anovulatory cycle**.
- **Pill Method (Fertility Control) :** Pills contain synthetic preparations of **Estrogen + Progesterone**.
- **Classification :** Classified into **3 types**:
 - a. Classical / Combined Pills
 - b. Sequential Pills
 - c. Minipills / Micropills

TYPES OF CONTRACEPTIVE PILLS

1. Classical / Combined Pills

- **Composition :—**
 - Moderate dose of **synthetic estrogen** (*Ethinyl estradiol or Mestranol*) +
 - Mild dose of **synthetic progesterone** (*Norethindrone or Norgestrel*).
- **Duration / Regimen :** Taken daily from the **5th to 25th day** of the menstrual cycle.
 - *Withdrawal of pills after the 25th day induces withdrawal (menstrual) bleeding.*
- **Mechanism of Action :—**

Continuous intake of combined pills



High circulating levels of Estrogen & Progesterone



Suppresses release of gonadotropins (LH & FSH) via negative feedback



Prevents follicular maturation and inhibits ovulation

- *Additional Progesterone Effect:* ↑ Increases viscosity/thickness of cervical mucus, making it hostile to sperm transport.

2. Sequential Pills

- **Composition :** High dose of estrogen + moderate dose of progesterone.
- **Duration / Regimen :—**
 - Estrogen alone taken daily for **15 days** (*from 5th to 20th day*).
 - Combined with progesterone during the **last 5 days** (*from 21st/23rd to 28th day*).
- **Action :** Primarily **prevents ovulation**.

3. Minipills / Micropills (Progestin-Only Pills)

- **Composition :** Low dose of **progesterone only** (estrogen-free).
- **Duration / Regimen :** Taken continuously **throughout the entire menstrual cycle**.
- **Mechanism of Action :—**

- Prevents pregnancy **without necessarily suppressing ovulation**.
- ↑ Increases thickness of cervical mucus → **inhibits sperm motility and transport**.
- Alters the endometrium → **prevents implantation** of the blastocyst.

COMPARATIVE SUMMARY OF PILL TYPES

Type of Pill	Hormonal Composition	Dosing Schedule	Primary Mode of Action
1. Combined Pills	Synthetic Estrogen + Synthetic Progestin	Days 5 to 25 of cycle (21 days on, 7 days off)	Suppresses FSH/LH surge → Inhibits ovulation + thickens cervical mucus
2. Sequential Pills	High-dose Estrogen followed by Estrogen + Progesterone	Estrogen for 15 days, then Progestin added for last 5 days	Inhibits ovulation
3. Minipills	Low-dose Progestin only	Daily continuously throughout the entire cycle	Thickens cervical mucus + prevents implantation (ovulation may continue)

DISADVANTAGES & ADVERSE EFFECTS

- a. **Compliance Issue** : Major practical difficulty is maintaining strict regular daily intake.
- b. **Medical Contraindications** : Not suitable for women with diabetes mellitus, cardiovascular diseases, or hepatic disorders.
- c. **Thromboembolism Risk** : Increased blood clotting tendency due to suppressed hepatic synthesis of physiological anticoagulants.
- d. **Cardiovascular Risks** : Hypertension and increased risk of myocardial infarction (heart attack).
- e. **Cerebrovascular Risk** : ↑ Increased risk of stroke.
- f. **Breast Pathologies** : Breast tenderness and potential ↑ increased risk of breast cancer.

LONG-TERM CONTRACEPTIVES (IMPLANTS)

- **Purpose** : Used to avoid the necessity of daily pill compliance.
- **Form** : Subdermal implants placed beneath the skin containing **progesterone / progestins**.
- **Mechanism & Efficacy** :—

Subdermal implant releases progestin slowly and continuously



Provides long-term reversible contraception for **4 to 5 years**



Common side effect: May produce amenorrhea or irregular spotting